

Frugality, A Positive Principle to Promote Sustainable Development

Damien Roiland¹

Accepted: 2 May 2016 / Published online: 10 May 2016
© Springer Science+Business Media Dordrecht 2016

Abstract Thinking and acting in favor of sustainable development is internationally recognized; it is necessary but societies and individuals are slow to adopt an appropriate behavior. International organizations such as World Commission on the Ethics of Scientific Knowledge and Technology recommend to opt for frugality, a principle emphasized to avoid over-consumption and consequently the depletion of natural resources. This article thus examines the principle of frugality by proving that it is not necessarily related to consumption as it is understood since the Enlightenment: we have to take into account the Asian perspective that changes the relationship between humankind and nature, first taking into consideration the good of the whole community. I argue that frugality is not necessarily in conflict with economy but certainly challenging it as expressed by the Indian terms *Jugaad* or *Gandhian*: a way of thinking and creating new projects that lead to fruitful solutions for the majority. Therefore I intend to provide a positive definition of frugality as a way of creating new projects that lead to fruitful solutions for the majority.

Keywords Frugality · Sustainable development · New behaviors · Innovation

Introduction

“He who will not economize will have to agonize”.¹ Today this saying attributed to Confucius in his *Analects*, seems to be addressed directly to the individuals and at the same time to our societies, especially to the ones who follow the principle of

¹ Wang and Lin (2009).

✉ Damien Roiland
damien.roiland@yahoo.com

¹ Ateneo de Manila University, Washington 40, Merville Park, 17000 Paranaque City, Philippines

mass consumption: naturally if we take care of the current and complex global system with the so called gap between the rich and the poor, Confucius' sentence could be reformulated or interpreted as follow: as the minority does not economize, the majority has to agonize. However the fact is obvious too: today industrial societies and people spend too much in goods and items when a majority only tries to survive. Consequences of non-renewable energy consumption for the future are disastrous. Problems about food have already been causing much concern for decades and the problems are getting worse: disparities and inequalities are increasing. Consequently sustainable development issues require the definition of principles shared by the whole community through a wide consensus such as the principles that the World Commission on the Ethics of Scientific Knowledge and Technology (COMEST) is currently working on.² Habits have to be changed in accordance with broad principles. One key principle could be the concept of frugality.³ A positive stance towards frugality is necessary for sustainable development: keeping the idea that frugality only reduces consumption is counterproductive. Giving a positive stance on frugality allows for the creation of a true relation between humans and nature where both are subjects and have the opportunity to flourish in harmony.

A Lively Tradition of Frugality in Asia

Countries where people have to face natural disasters or disorders (limited resources, cataclysms, overly abundant population...), have traditionally stressed the importance of frugality: Jainism, Buddhism, Daoism and Confucianism give specific attention to nature and collective concerns. Who knows if it is the religions or the social organization that brings out the importance of frugality in Asia? The facts show that this principle has been part of the culture for a long time. Religion stresses the importance of living in harmony with nature. Frugal life is a natural and social necessity; it is the path to serenity and happiness. A moral life of material simplicity is virtuous as well. This responds to an early assessment: a large population has to share available goods. It also highlights a dangerous individual trend, greed. Energy's sources are no longer free but some individuals keep them. Between individual greed, and over-consumption, frugal life becomes a good path. Rudyard Kipling in *The Second Jungle Book* lets Western readers discover the story of these many men who become monks after having led a major career in politics.⁴ Kipling presents the life and the miracle of the monk Purun Baghat previously

² COMEST (2005).

³ "By 2015, COMEST envisages to produce a document that combines the principles and responsibilities already stated by the Commission in 2011. (...) COMEST has already identified a number of principles of relevance for a comprehensive ethical framework in relation to climate change, such as the principles of sustainability; global justice; solidarity of life on earth; resilience; frugality; biological and cultural diversity; integrity as applied to ecosystems; and safeguarding and promoting the interests of present and future generations." COMEST, "Briefing about the work of UNESCO's COMEST on a general framework of ethical principles for climate change", 4.

⁴ Kipling (1906).

known as Sir Purun Dass K.C.I.E, Prime Minister of an Indian State. At the end of the story, the holy man woken up by his friends, wild animals, has just enough the time to save the sleeping villagers from a massive landslide. It is surprising to see how men and women in India can decide to lead a simple life, walking the path as pilgrims or choosing a secluded monastery in the mountains to share a frugal life with villagers. Stories about monks and religious men and women are common in Asia. These people play an important role in society: they are part of the community life, well respected and followed as guides. One of the oldest religions in the world (seventh century BC), Jainism, was widely adopted in India during long periods.⁵ This religion appears in the Ganges basin; it emphasizes spiritual independence and equality between all living entities on earth. It prescribes a path of nonviolence and self-control that are the medium through which everybody becomes able to be free from repeated rebirth. Vardhamana Mahavira is one of the first historically known religious. This monk and teacher is probably a contemporary of Buddha. Vardhamana Vira and Buddha are proposing a path of asceticism with a deep respect of nature.

The different traditions of Buddhism are highlighting the importance of frugality. At least two points are interesting: the man's relationship with his own self and with cosmos, the emphasis given to two material goods. Buddhists remind us that attachment to property leads to constant dissatisfaction.⁶ Current lifestyles, compartmentalized, associated with the desires of goods, ever lead to more desires and dissatisfactions. Happiness cannot be obtained. The effort must relate to the detachment from material goods, at least to reposition priorities and thus recognize how little material goods helps to self-fulfillment. This self fulfillment has to be inquired by each individual. But Buddhists have very particular perception of self: the individual is not the center of himself. But the cosmos appears as the center of each individual. The self is thus a search for harmony with the cosmos. Therefore frugality helps to position the value of material goods as it should be. Nature is respected as an integral part of the cosmos, to which each individual wishes to belong.

Daoists also give different levels to the idea of frugality. Lao Tsu explains:

"I have three precious things, which I hold fast and prize. The first is gentleness; the second is frugality; the third is humility (...). Be gentle and you can be held; be frugal and you can be liberal; avoid putting yourself before others and you can become a leader among men".⁷

Harmony is the key to explain the relationship between humans and nature, which is considered as sacred. This harmony allows energies to flow, especially wind and water.⁸ The human body is a microcosmic image of the macrocosmic nature that enfolds them. Living in nature (especially in mountains and caves) gives the possibility to amplify the body's energy system. This does not mean that all Daoists

⁵ Sangave (2006).

⁶ Zsionai (2008), p. 281.

⁷ Tsu (1905), p. 35.

⁸ Miler (2008).

have to avoid cities. But they consider nature as a great vehicle to go across another realm, to attend a higher transcendence. Daoists highlight sober or moderate life because their aim is to reach immortal life. Mortal objects like goods or food are not the ones we have to accumulate because they cannot be the ones that feed body and soul. Both Daoism and Confucianism pursue harmony, which can be obtained through different ways. Daoists look for a frugal life related to nature, when Confucianists look for a frugal life in relation to society.⁹ “He who will not economize will have to agonize”. Therefore the double Chinese sign used to express the frugality 节俭-Chog shes- insists on the economy, the non-wastefulness with the first sign while the second sign is related to the sobriety.

Through this short approach of frugality in Asia, we can summarize the idea that this principle is a cultural, religious rooted concept: by frugality we open a path of happiness to use for oneself what is needed, to share with others the benefits, to facilitate the social life and to establish and balance a relationship with the whole universe including nature. Frugality is also an important way to let our transcendence concerns evolve. In other words, the aim is to avoid focusing or to not be focused on the materiality that conducts toward insatiability and dissatisfaction. People who want to be autonomous and develop their spirituality, people who take care of the whole community consider frugality as a necessity. Being frugal, being sober points ‘being’ instead of ‘having’ out. Certainly there are different possible interpretations about the needs each individual has to acquire for his or her own. And the different religions, philosophies, shortly presented her, are more or less austere. The importance of nature in this relation is also interpreted in various ways. If the way to talk about nature today is very different from what it used be, that means the texts we study cannot provide a direct answer to our new concern of sustainable development. Asian traditions we have just studied provide a positive approach to frugality. It is a path of self-development respectful of others and nature. Each individual always looks for harmony. This is not so clearly the point in European contemporary culture. And a simple presentation of the evolution of frugality’s perception from antiquity to the Enlightenment demonstrates it.

Frugality, the Negative Paradigm Shift Highlighted by Diderot’s Encyclopedia

Frugality becomes an outdated idea during the Enlightenment. The superiority of man over nature, the power of the individual vis-à-vis the group, the development of consumption, these emerging ideas leave out what frugality could bring. The short article about frugality in the *Encyclopédie* of Diderot proves it. What’s happened? Reason makes us “masters and possessors of nature” according to Descartes.¹⁰ Due to the fact that nature is not firstly perceived as a subject, a partner, however as an obedient object, nature has to be submitted to the human will. In the Western dominant tradition, the relationship between man and nature is unequal. The

⁹ Weiming (1998).

¹⁰ René Descartes, *Discourse on the Method*, Tome 1, part 6.

primacy on having can be seen with liberalism, capitalism and the economic system. Producing, keeping, accumulating... These ideas are spread all around the world, including in Asia. Confucius' motherland is the one that produces the most today. Chinese capitalism model spreads its products all around the world and dangerously develops its own domestic market. If a big part of the Chinese population is still frugal, conversion to the mass-consumption is just a question of time: everyone dreams of having all the new gadgets. Again, this acceleration of mass production and mass consumption is disastrous for sustainable development. The capitalist economic model is an international reference. And people have now a negative perception of frugality.

From the Western tradition, frugality comes from the Latin word "frugalis" which means that produces. The frugality was early interpreted as thrifty, sober during the classical period. The sentence "Esse frugi bonae" translated by "to be capable of giving a good crop" was then applied to a person of morality (Plautus). If frugality is considered as a virtue and encouraged by the different Greek schools of thought, Diogenes and the Cynics are the only ones who stand for a radical ascetism an example of it being Jainism in India. Here frugality is an end in itself. Most of the Latin and Greek philosophers advocate a free choice of the will to moderation. It is as a medium to obtain the harmony that can be individual or collective: the harmony in the City (Plato), the personal harmony (Epictetus, Epicure)... Christianity like the two other religions of the Book clarifies and structures the frugality for the whole community: frugality of life is an obligation for the ultimate path of resurrection. For example in the *Summa Theologiae* Thomas Aquinas highlights seven virtues, including temperance. And with two of the seven deadly sins, gluttony and greed (as the accumulation of goods for themselves), humans are clearly obliged to moderation.¹¹

During the Enlightenment period, the religious perception of frugality started to disappear. It is analyzed in relation with consumption. The article on frugality in the *Encyclopedie Universelle* makes the paradigm shift concrete. "This virtue does not concern only the table, it is about habits whose frugality is the firmer support (...). In our countries of ostentation and vanity, frugality has lots of difficulties to keep an estimable rank".¹² The author of this article, Louis de Jaucourt, revives the antique definition of frugality: the search for frugality has a double advantage: it offers individuals the joy of simplicity, the ability to observe the religious commandments; it enables the State, following the example of the ancient cities, to accumulate individual superfluous and with it to carry out its major projects for public good. However the author also analyzed frugality in opposition with consumption, a way of life where the desire part is prominent. Frugality is not up with current tastes that prefer only the song of fame and luxury the sirens while frugality is henceforth associated with avarice. The art of the ancients able to live with the same moderation, simple life and honors, does not seem to beckon to collective imagination. That is the reason why the article sounds like a complaint: How to love frugality? It seems impossible as references remain the fortune of those who have

¹¹ Nash (1995).

¹² Diderot (1757).

“succeeded”. As noted by Montesquieu, corrupt beings through pleasures cannot be connected to simplicity. What makes oneself miserable is not poverty but failure of succeeding for those who want to be rich. Frugality loses its spiritual values. And it no longer participates in the link that can unite individuals for the greater good of the community.

The frugality tradition in Western societies is mostly hidden by other concerns. Importance is given to material wellbeing, money, upward mobility and social success, health... Diderot and the authors of the *Encyclopédie* witness to a paradigm shift. Frugality is not a virtue between extravagance and avarice anymore. It becomes impossible for individuals corrupted by pleasures to be attached to simplicity. People seem to have a negative perception of frugality.

Reviving the Relationship Between Human and Nature; Pierre Rabhi's Approach

Frugality and nature today appear under new perspectives: faced with major economic and environmental crisis, individuals are starting to react. They veer between an obliged frugality and one they have chosen. Individuals start to have a less negative perception on frugality: sociologists observe this. And especially new personalities sensitive to the problem of sustainable development are heard. The example of Pierre Rabhi is particularly relevant. This reaction is still negligible: the stirring of conscience only appears with the example of few men, in international conventions. And the results are not yet grounded in important global effects. However the actual economic crisis increases the need to find new frugal behaviors and strategies even (especially) in Western and developed countries.

Sociologists from CREDOC present new consumption ways from the middle class to answer to the new aspirations of well-being including sharing, social link and sustainable development.

Since the 2008 crisis, frugality is being replaced on the reflection's frontline concerning on the way we consume. The consumer is being realized that the poor quality is more expensive in the long run. Frugality is today reducing the quantities consumed while focusing more on quality products. In other words, this is about to use less in order to consume better.¹³

The priority is given to quality of life, realization of the self, psychology and questions concerning the meaning of life. Therefore for 20 years, the sensibility to frugality has been reappearing in a new light. Henceforth it is presented to cope with society consumption. Frugality thus finds a more concrete and accurate field of application. And the means used to practice frugality will be diversified while the aims to be obtained are many, varied and less sectarian. Few active minorities have emerged and have kept on growing. They come from political and philosophical movements which rebel against mass consumption, sensible to environmental problems, anew asserting the search for individual and social harmony. Then

¹³ Siounadan et al. (2013), p. 21.

frugality is not experienced as a constraint due to the economic crises, but as the result of a choice. It is about putting people at the heart of concerns, privileging the quality of life, relationships as exchanges rather than increasing blindness the quantity. This way of living is named sometimes “slow attitude” in reaction to the acceleration, the contraction and primacy of the time. For others it is one of the reasons for the growing attraction for Eastern wisdoms/religions. And for others it is the motivation for ecological commitment. Although these pathways are not necessarily differentiated, some individuals choose a voluntary deceleration to highlight sharing, the importance of exchanges. Globalization takes a new negative meaning. It becomes de-soilation and desolation. For too long many individuals have not taken care of Mother Earth. Too much exploitation has exhausted the soil. And today, world citizens have to recreate a dialog with Mother Earth. Opened to the whole world, they also restart communities, local relationships to promote micro-economy, good relations among neighbors. Therefore the proposed consumption ways have become more respectful of nature and a post materialistic approach is sought.

Some experiences show us the danger and the irony of the actual model of consumption. Pierre Rabhi¹⁴ explains a famous one used by few European politicians and ecologists: during the 80 s a truck transporting tomatoes from Holland to Spain had an accident on a French highway: it crashed into another truck that was transporting tomatoes from Spain to Holland.¹⁵ The laws of market become senseless. This story highlights the grant’s perversion, the blind consumption of energies. Therefore economic growth (and maybe its toned down version of sustainable development) can be the problem and not the solution today. The goods from the Earth become consumer goods subjected to the market laws. They turn into products available at any season, moved all around the world. Transformed to be made intensively, food lost its nutritional and poetic skills... The economic model turns into a destructive system. Consumers are rediscovering the benefits of frugality for their wallet first, reacting to problems of sustainable development too. And increasingly they are convinced by common sense reflections from simple and happy men. The example of Pierre Rabhi, a self-taught man who ran for the French presidential elections in 2002 is significant. Frugality, previously perceived as negative, restores its respectability.

¹⁴ In France, Pierre Rabhi is a famous farmer and writer. Self taught man, he is born in Algeria in 1938. He starts to work in a Parisian factory. A few years later (1959) he and his wife buy a land in a poor French province. His aim is to conciliate the human history with a natural balance. Engaged in an agricultural ecology center, he gives his expertise to young French farmers, to farmers in the Sahel desert and then in several countries in Africa, Middle-east and Eastern-Europe. At the request of the U.N. he prepares in 1998 some concrete proposals for the implementation of his plan for the Agreement on action against desertification (CCD).

¹⁵ Rabhi (2008), p. 91.

The Coming Back of Frugality in Western Societies: Pierre Rabhi's Approach

Pierre Rabhi invites us “to join our forces to serve and promote simple values like benevolence with regard to the ones who surround us, a sober life to let other living, the compassion, the solidarity, the respect and the conservation of Life in any form whatsoever”.¹⁶ How can Pierre Rabhi's gives a fruitful approach of frugality? Which arguments justify the return to the land according to him? In the afterword of *Manifesto for the Earth and Humanism (Manifeste pour la terre et l'humanisme)*, a concrete process made by the author and experts presents how the production of organic goods regenerates the soil, renews bio-diversity, gives a good quality and quantity and avoids pesticides and other pollutions. According to the social and economics' point of views, this approach of a local and organic production brings a viable and helpful shape. “It is easier to imagine how the society can mute and in which way we can have more power on our future, individually and collectively”.¹⁷ Therefore the Pierre Rabhi's main arguments are to come back to the earth to cultivate it and be nourished in return, to pursue a simple life that is source of happiness. We also have to break with the agro-industrial system: the quality of the products is not healthy; the food economy is included inside a more global economy. We have to recreate local social networks. Pierre Rabhi proposes to change the logic that governs us. A paradigm shift is advocated from his wishes: putting human and nature at the heart of our concerns. Frugality is situated among the keys of this paradigm shift:

Considering moderation, frugality as an art to be in harmony with oneself, with the others and with the nature. It is a calling into judicious question about “progress”, a conscious act of liberation of the obsession of lack that generates anxiety, violence and intolerable injustice.¹⁸

Pierre Rabhi emphasizes the idea of moving out from the actual agriculture and economical system. He highlights the concepts of responsibility and frugality. According to him, one of the most essential triggering factors to bring about a salutary paradigm shift is the sudden rise of the petrol barrel. Added to the transportation cost to dispatch the goods to the different markets, the production cost with the actual conventional system will blow up in the European Union like in the U.S.A.: it is necessary to spend about 2.5 metric tons of gas to produce a metric ton of fertilizer. In the E.U. The grant system to industrial and chemical single-crop farming is contested and attacked by the financial field and by the ecologists. And then population is regularly informed by food-processing industry scandal or other comical story like the one displayed *supra*. The mistrust increases.

¹⁶ Pierre Rabhi, *ibid.* p. 93.

¹⁷ Rabhi (2008), p. 118.

¹⁸ *Ibid* p. 98.

Through his foundation¹⁹ Pierre Rabhi has been calling for a return to the earth with an agriculture that respects the soil. 30 % of arable land have been sterilized or transformed in laterite by intensive practices over the last 30 years. The humus, this so thin skin of the earth is the first one that nourishes the whole humanity (products from the sea also are a non-negligible part for healthy diet). Therefore Pierre Rabhi is asking the farmers to nourish humanity, obviously, and also to nourish the soil itself. This double movement is the condition to create a virtuous circle: the soil needs to be regenerated; farmers have to take care of it and even to cure the soil to compensate natural erosion, the exploitation and especially the results of the intensive industrial exploitation. The quality of the food depends on the quality of the humus and then of the growing process. The farmer who has pulled benefits from the soil lets it then rest so it can regenerate itself. Simple technics have to be shared to (re)create “oasis in any place”. The soil clearly becomes the skin of the planet, thin, fragile and essential. The personality and the health of individuals as well as the one of the earth can be discovered from the skin/soil: it is the constant mediator between the inside and the outside, an area of exchanges and breathing. Here is the manifestation of the man inwardness and the world that encompasses him. The analogy can be pursued with the psychoanalyst Didier Anzieu, *The Self-skin*:²⁰ we discover the fruitful relations between the earth, the soil/skin and the self. Each is the witness of the health of the other; each one is essential to the other and nourishes him/it.

Around Pierre Rabhi, citizens choose to take responsibility for food production. They become attentive to its origin, its method of cultivation, its quality... In this way they create a direct relationship with the farmer by paying him his crop in advance to share the risk, to avoid price fluctuations. Each week the farmer meets them at the market to submit his production. This is an economic and social system proposed of course with a primary emphasis on ecology: water pollution, diversity of lifestyles and regions, removal of chemical fertilizers, biodiversity conservation, seed conservation and promotion of heritage and indigenous practices (Rabhi 2014).

Making the choice of a new agriculture handling directly related consumers and farmers might seem trivial. Yet Pierre Rabhi like many environmentalists around the world considers that this is the beginning of a profound change in the global economy, therefore a change in lifestyle. Pierre Rabhi and the Hummingbird Movement expresses the will to first of all change individual minds in order to push for institutional changes. On the one hand the action of few individuals has no important effect on the global scale but on the other hand this action is a beginning.

¹⁹ In 2006, the Movement for Earth and Humanism (Mouvement pour la Terre et l'Humanisme) created in 2002 for the presidential election in France becomes Hummingbird Movement (Association Colibri). The movement's choice of name comes from the favorite legend of Pierre Rabhi. This legend also inspired the founder of the Green belt movement, Wangari Maathai, Kenyan environmental and politician activist, Nobel peace prize laureate. This South American story narrates that a day a huge wildfire was devastating the primal forest. Disconcerted, the entire animals were watching the disaster happening. The hummingbird was active for its part. He was carrying few drops of water in his beak to throw them on the fire. An armadillo annoyed by this pathetic agitation told him: are you insane, hummingbird? Do you believe that you will truly extinguish the wildfire with these drops? And the hummingbird to answer: I do my fair share of the work.

²⁰ Anzieu (1995).

To induce a paradigm shift that will offer a better quality of life for everyone, we need first to start by ourselves. With the hummingbird legend (cf. note 13), Pierre Rabhi invites to do simple actions to live happily with frugality.

To Strengthen a Positive Definition of Frugality

Can we consider frugality to be a virtue?²¹ This approach holds an obvious interest, would this be only as a horizon or as a yardstick.²² And based on this guideline, it is possible to provide directions that may achieve a broad consensus supported both by individuals and especially by companies. While there may still be some elements to review, the main thing here is to make the establishment of rational purpose to all those who are longing to live well possible. In other words can we consider virtuous these traits that allow individual autonomy and life in harmony within a community and an environment so that each and every being can flourish? But in fact, the COMEST does not directly fit into the debate about virtue by avoiding the use of this word and by preferring to call it principle.²³ “When ethics is integrated into political and legal thinking, one should be careful to (...) strive for as much practical consensus on moral judgment as possible. One way to do this is to deflect our thinking away from ethical theory and rather focus on more specific ethical principles and ethical values” (COMEST, 17). What is first targeted, also appears here, namely to define the general principles needed for an ethical platform that is used to form agreements on which each community, State, can make a clear commitment.

Is this research on frugality only a problem of the rich and thus one that has no relevance to the vast majority of the population? Obviously, what raises these questions today is related to overconsumption in the wealthiest.²⁴ This does not seem so obvious to others: “A motivation to save the planet and being frugal are found unrelated”.²⁵ However, as presented below, frugality can be perceived as a reaction to the problem of bad/over-consumption concerning balance problems, recycling (Rabhi-Credoc). That said, all of us remain concerned by chosen and mastered consumption, in harmony with nature. Everyone, rich or poor, every society, whether they be developed or developing must be sensitive to these issues. And frugality can lead to a better distribution of wealth as well as to a redefinition of the ends sought by the communities as individuals.

Could the world system as it works today, withstand a drop in consumption itself driven by the advent of frugality followed more generally? However this more economical than philosophical question cannot be ruled out here. Most movements that advocate frugal lifestyle are opposed to consumer society as well (Rabhi, “slow attitude”). Yet this opposition developed between frugality and consumer society

²¹ Griffin (1998).

²² Gambrel and Cafaro (2010), p. 88.

²³ COMEST (2014).

²⁴ Pepper et al. (2009).

²⁵ Lastovicka et al. (1999), p. 88.

does not necessarily appear as obvious. There are certainly important and necessary changes. And that does not necessarily lead to destroy the existing liberal model. The approach made by Navi Radjou is particularly interesting. It allows us to redeploy the notion of frugality from the Asian perspective with a particular concern for harmony with nature, integration in the community. And it starts to give an answer to these types of economical critic.

Innovative Frugality for Global and Individual Changes

Frugality embodies a positive principle in major ways: it is possible to be frugal and effective and to reconcile economic growth and environmental preservation. Rather than waging a battle against over-consumption, individuals develop a new vision of frugal behavior based on a different perception of innovation. To be frugal is unmotivating in itself for most of us. Therefore is it a vain effort to emphasize the importance, the value of a frugal behavior, if we consider the little effect on the whole? Is it even counterproductive if we consider how growth is a key factor in the economy today and also certainly tomorrow? The liberal-capitalist model is always reinforced and the communist nation of China became in the end of 2014 the first world economic power... Frugal habits in industrial societies are conducted by the conjunction of the actual economy situation and the different ecological movements. The idea of a sustainable development increases. More and more individual actions are made and at the same time international congresses about alimentation, water preservation, forest protection, animal's protection and global warming succeed one another. Pragmatic results are not visible (yet) to the point of being dubious with the results being common. We have to admit that the media are focused on international actions about an important but distant problem, global warming. However close or immediate tragedies are not clearly echoed in the press and in the international community. Whether they be starvations or related to water access and waste processing, most of them have not received appropriate solutions yet. And as far as individuals are concerned, the habits of those who consume and pollute a lot are only changing very slowly.

However frugality can be understood in different ways. From a negative perception, going back to the Eastern cultures is fruitful. The Indian perception is particularly interesting and relevant here. On the level of production and innovation frugality is giving important keys: it is possible to be frugal and effective, to bring together economic growth in view of ecology. Today the manager has to face a challenge flow in a world where everything changes. This is a world of scarcity, environmental issues, interconnection, complexity and diversity. Also frugality is an aspect which is now recognized among managers in terms of innovation. "This is an optimistic stance," considers Navi Radjou in his book *Jugaad Innovation: Think Frugal, Be Flexible, and Generate Breakthrough Growth*. The outline of this book is a definition of Jugaad in itself. Six principles are highlighted: Seek opportunity in adversity/Do more with less/Think and act flexibly/Keep it simple/Include the margin/Follow your heart. "Jugaad is, quite simply, a unique way of thinking and acting in response to challenges; it is the gusty art of spotting opportunities in the

most adverse circumstances and resourcefully improvising solutions using simple means. Jugaad is about *doing more with less*.²⁶ Therefore the importance of frugality does not appear as opposed to economic growth, a kind of programmed economic decay like European environmentalists are strongly considering. Jugaad is more an invitation to a different frugal life: the search for more efficient energy waste, more effective invention, more opened in management. Some confidence is attributed to man to adapt himself to his new environment and face the challenges of energy limitations as increased pollution.

The Hindi words “Gandhian” or “Jugaad” are very popular. Indians use them to express an innovative fix, an improvised solution born from ingenuity and cleverness. “Gandhian engineering” was popularized with the citation of Gandhi:

I would prize every invention of science made for the benefit of all. There is a difference between invention and invention. I should not care for the asphyxiating gases capable of killing masses of men at time. The heavy machinery for work of public utility which cannot be undertaken by human labour has inevitable place, but all that would be owned by the State and used entirely for the benefit of the people. I can have no consideration for machinery which is meant either to enrich the few at the expense of the many or without cause to displace the useful labour of many.²⁷

With Gandhi the aim is obvious: he was always wary of the use of machinery but he considered machinery as a necessity, especially for the benefit of the whole. According to him the danger was the use of machinery and not the machinery itself. Machinery and inventions do not have to be based on greed and cannot justify exploitation. However, inventions that can improve the condition of the mass (and not only a small group) are always welcome. In other words frugal inventions are done to the public sphere and all the members, including the poorest are complete beneficiaries.²⁸

India becomes a pioneer of a new concept of innovation. It is rooted in its tradition we highlighted as *supra*. When millions of people do not have enough to live decently, it is necessary to find alternative solutions to face lots of problems. Today, over 40 % of Indians are unbanked, off the electricity grid. They have limited access to healthcare and education. Four billion people in the world have less than 9 USD per day, 2 billion have between 9 and 55 USD and only 0.5 billion have more than 55 USD (World Resources Institute). These billions of poor people can become a key target for those multinational companies that want to sell on the global scale, provided that they do not intend to make a high-margin benefit on the product but try to interest the biggest possible number of consumers: the product must be simple, effective and as cheap as possible. The term used to explain this way of innovation can be Jugaad. This Hindi word generically used can be translated by “bring together”. Jugaad is proposing alternative solutions, fixing up

²⁶ Radjou et al. (2012), p. 4.

²⁷ Gandhi (1935).

²⁸ Platt (2012). <http://theinstitute.ieee.org/career-and-education/career-guidance/introducing-gandhian-engineering573>.

temporarily things. It solves the current situation with simple and effective solutions. Poor people have no other choice than to create by themselves fridge made in clay without electricity, sending miscalls codes to communicate for free, using Coca Cola plastic bottle as storages or roof lights, etc. The term Jugaad became famous to the English speakers with the presentation of a tractor made by Indians farmers from hinterland's villages. These poor people could not buy a tractor. They made it by combining a chassis of a trolley with a diesel water pump. They attached it with a steering rod and a wheel. Then the barrel of a drum was used as a fuel tank. Jugaad word has equivalent expression in many languages; for example Gambiarra in Brazil, Kanju in Africa, Jiejian Chuangxin in China, DIY in the US, Système D in France, Para-paraan in the Philippines. Today the Internet overflows with Jugaad pictures from India, China, Brazil or Kenya... It is certainly amusing but it demonstrates also a great capacity of adaptation with mere resources. This is very attractive for everyone: for the poor, for the one who face a sudden difficulty but also for the ones who want to perform. Combining Jugaad with long term plan within an international company is possible. Carlos Ghosn, the joint chief of the two cars companies Nissan (Japan) and Renault (France) stated in 2006 in front of media "frugal engineering is achieving more with fewer resources. The cost of developing a product in the West is high since engineers there use more expensive tools. In India, they achieve a lot more with fewer resources". This analysis of the frugal engineering in India leads the chief director of Nissan and Renault to create three joint venture companies with Indian partners. International companies are more and more attracted by Jugaad innovations to produce cheaper brand. The Tata Nano car and the Nokia 1100 are the best-known results in the world.

Everyone must be able to benefit from the proposed innovations. An orientation towards growth respecting nature and reducing inequality from above through efficient and universal innovations has now taken place. The ubiquitous cyclical pattern in Asia becomes a reference and replaces the linear growth. Geological cycles of water, rocks of living matter leads to innovate to a recycling society: water recycling, raw materials, waste such as food. The frugal ecological revolution can be understood as the astral model instead of the French historical reference: the revolution is to give back, to create cycles.

Conclusion

Frugality is a concept that forces to redefine our priorities in life, in economy in a global system. An Ethic of frugality on different levels appears essential to face the actual challenges. However frugality cannot be presented in opposition with creativity and economic growth. On the contrary it is a fruitful way to take advantage the opportunities such as the ones that the Indian word Jugaad encompasses.

Through its socio-historical dynamics, frugality obviously fluctuates between choice and compulsion, between the individual and the collective aim. Beyond these developments, frugality seems to be a value that is resisting the ordeal of time while transforming itself at the whim of different representations of happiness. With

Confucianism, with the emerging philosophies during antiquity in Europe, frugality is a path to find the truth and the harmony of the city. Most Asian wisdoms/religions (Jainism, Buddhism and Daoism), and in the West the growing influence of religion during the Middle Ages led to a search for soul salvation through standardized practices. During the Enlightenment years, frugality seemed to be restricted to a secular and earthly purpose, then with the emergence of consumerism it was relegated to the background. With Pierre Rabhi frugality came back to the front stage. It was considered a way of resistance. And now, frugality embodies a positive principle in major ways: it is possible to be frugal and effective and to reconcile economic growth and environmental preservation. Rather than waging a battle against over-consumption, individuals develop a new vision of frugal behavior based on a different perception of innovation. Frugality is also understood as a powerful innovation lever leading to the development of new, simple and effective technologies for the greatest number. It also comes to reconnect practices called frugal with more natural and spiritual aspirations. And the model of Asian wisdom is then a great source of inspiration.

And to build a bridge between different cultures, we can end this essay with a reference set in the middle-east, precisely in Turkey:²⁹ After a journey around the world, from Westphalia to the New World, from the beautiful gardens of an old castle to the magnificence of the Eldorado, Constantinople is the place where one can do anything. “Let us cultivate our garden,” answers Candide to Pangloss at the end of Voltaire’s famous philosophical tale. Although in French, “cultiver son jardin” refers to the knowledge “la culture”, we are used to interpret the last sentence of Candide as an invitation to think and meditate. Nonetheless we can also keep the literal meaning of the expression and be delighted by the “preserved citrons and pistachio nuts”. The movement back to the land yields lots of fruits: giving food and thoughts while enjoying our own production, re-discovering simple happiness inside a community. It seems that this kind of wisdom has disappeared from Western philosophy since Voltaire until today. However in Asia, with looking for human’s well-being means looking for harmony between humans and nature; thus highlighting the importance of frugality for a more rooted way of life.

References

- Anzieu, D. (1995). *Le moi-peau*. Paris: Dunod.
- COMEST. (2005). *The precautionary principle*. Paris: UNESCO.
- COMEST. (2014). *Briefing about the work of UNESCO’s COMEST on a general framework of ethical principles for climate change*. Paris, UNESCO, December 4, 2014. <http://unesdoc.unesco.org/images/0023/002310/231082E.pdf>.
- Diderot, D. (1757). “Frugalité”, *Encyclopédie*, Paris, Briasson
- Gambrel, J. C., & Cafaro, P. (2010). The virtue of simplicity. *Journal of Agricultural and Environmental Ethics*, 23, 85–108.
- Gandhi, M. K. (1935). *Young India*. Bombay, June 22 1935.

²⁹ Voltaire (1918).

- Griffin, J. (1998). *Is frugality a virtue? If not an end in itself, frugality can lead to a better quality life* (Vol. 51). Paris: UNESCO Courier.
- Kipling, R. (1906). *The Second Jungle Book*. New York: The Century Co.
- Lastovicka, J. L., et al. (1999). Lifestyle of the tight and frugal: Theory and measurement. *Journal of Consumer Research*, 26, 85–98.
- Miler, J. (2008). *Daoism*. Oxford: Oneworld publications.
- Nash, J. A. (1995). Toward the revival and reform of the subversive virtue: Frugality. *The Annual of the Society of Christian Ethics*, 15, 137–160.
- Pepper, M., Jackson, T., & Uzzell, D. (2009). An examination of the values that motivate socially conscious and frugal consumer. *International Journal of Consumer Studies*, 33(2), 126–136.
- Platt, J. (2012). Introducing Gandhian Engineering. *The Institute*, IEEE, February 2012.
- Rabhi, P. (2008). *Manifeste pour la terre et l'humanisme*. Paris: Actes Sud.
- Rabhi, P. (2014). *Ce que nous dit la nature*. Paris: Les éditions du Relie.
- Radjou, N., Prabhu, J., & Ahuja, S. (2012). *Jugaad innovation*. San Francisco: Jossey-Bass.
- Sangave, V. A. (2006). *Aspects of Jaina religion*. New Delhi: Bharatiya Jnanpith.
- Siounadan, N., Hébel, P., & Colin, J. (2013). *Va-t-on vers une frugalité choisie?* Paris, CREDOC, no. 302.
- Tsu, L. (1905). *The sayings of Lao Tzu* (Lionel Giles, trad.). New York: E. P. Dutton.
- Voltaire. (1918). *Candide* (p. 918). New York: Boni and Liveright Inc.
- Wang, C. L., & Lin, X. (2009). Migration of Chinese consumption values: Traditions, modernization and cultural renaissance. *Journal of Business Ethics*, 88, 399–409.
- Weiming, T. (1998). Beyond the enlightenment mentality. In M. E. Tucker & J. Berthrong (Eds.), *Confucianism and ecology: The interrelation of heaven, earth, and humans*. Cambridge: Harvard University Center for the Study of World Religions.
- Zslonai, L. (2008). Buddhist economic strategy. In *Frugality: Rebalancing material and spiritual values in economic life* (p. 281). Oxford: Peter Lang Academic Publisher.